1. Complex Numbers

1.1. The Basics

$$i^2 = -1$$

- General form for all complex numbers:

$$a + bi$$

- From this we say:

$\mathrm{Re}\,(a+bi) = a$ & $\mathrm{Im}\,(a+bi) = b$

- Conjugates:
  - The complex number z and its conjugate z^*

$z = a + bi$ & $z^* = a - bi$

- Arithmetic:
  - **Addition** and **Subtraction**: add and subtract real and imaginary parts with each other
  - **Multiplication:** carry out algebraic expansion, if i^2 present convert to -1
  - **Division:** rationalize denominator by multiplying conjugate pair
  - **Equivalence:** equate coefficients

1.2. Quadratic

- Use the quadratic formula:
  - $b^2 - 4ac$ is a negative value
  - Pull out a negative and replace with i^2
  - Simplify to general form
- Use sum of 2 squares: consider the example

Example:

Solve: $z^2 + 4z + 13 = 0$

Solution:

Convert to completed square form:

$$(z+2)^2 + 9 = 0$$

Utilize i^2 as -1 to make it difference of 2 squares:

$$(z+2)^2 - 9i^2 = 0$$

Proceed with general difference of 2 squares method:

$$(z + 2 + 3i)(z + 2 - 3i) = 0$$

$$z = -2 + 3i \quad and \quad z = -2 - 3i$$

1.3. Square Roots

Example:

Find square roots of: $4 + 3i$

Solution:

We can say that:

$$\sqrt{4+3i} = a + bi$$

Square both sides

$$a^2 - b^2 + 2abi = 4 + 3i$$

Equate real and imaginary parts

$$a^2 - b^2 = 4 \qquad 2ab = 3$$

Solve simultaneous equation:

$$a = \frac{3\sqrt{2}}{2} \qquad b = \frac{\sqrt{2}}{2}$$

$$\therefore \sqrt{4+3i} = \frac{3\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i \qquad or \qquad -\frac{3\sqrt{2}}{2} - \frac{\sqrt{2}}{2}i$$

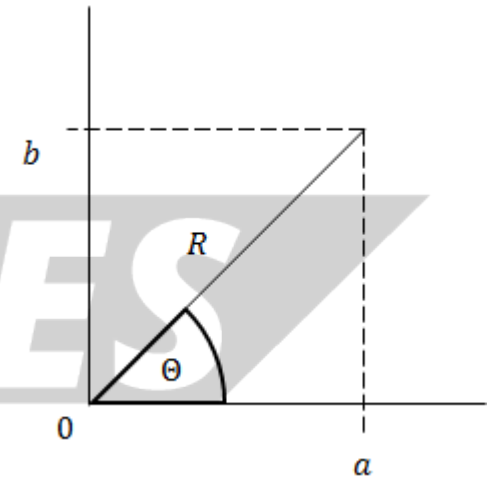
1.4. Argand Diagram

For the complex number $z = a + bi$

$$|z| = \sqrt{a^2 + b^2}$$

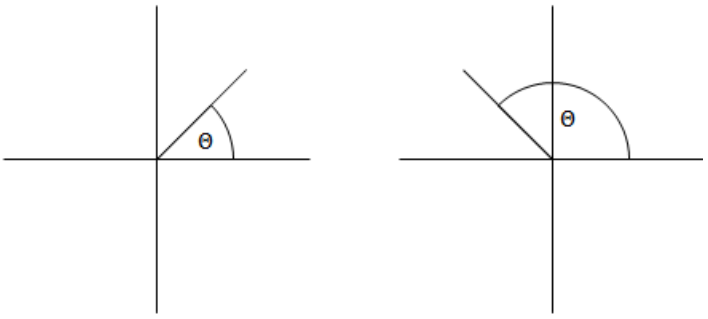
$$\arg z = \tan^{-1}\frac{b}{a}$$

- Simply plot imaginary (y-axis) against real (x-axis):



Argument:

Always: $-\pi < \theta < \pi$



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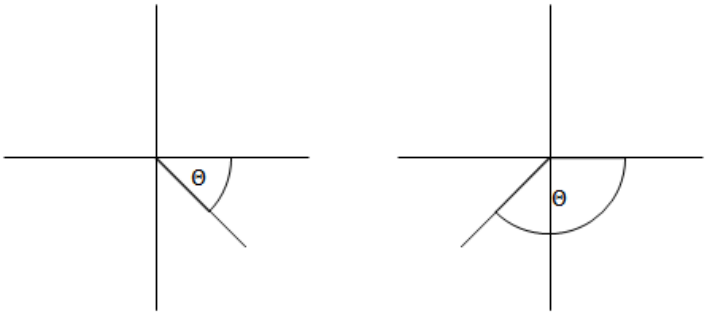
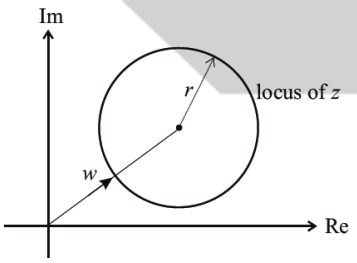


- The position of z^* is a reflection in the x-axis of z

1.5. Locus

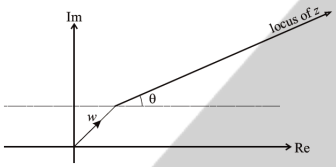
$$|\mathbf{z}-\mathbf{w}|=\mathbf{r}$$

The locus of a point z such that $|z-w|=r$, is a circle with its centre at w and with radius r.



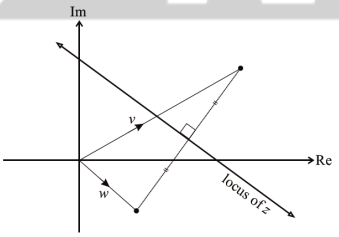
$$\arg(\mathbf{z}-\mathbf{w})=\theta$$

The locus of a point z such that $\arg(z-w)=\theta$ is a ray from w, making an angle θ with the positive real axis.



$$|z-w|=|z-v|$$

The locus of a point z such that $|z-w|=|z-v|$ is the perpendicular bisector of the line joining w and v



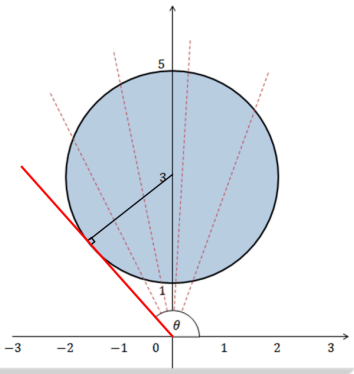
{W11-P31} Question 10:

On a sketch of an Argand diagram, shade the region whose points represent the complex numbers z which satisfy the inequality $|z-3i|\le 2$. Find the greatest value of $\arg z$ for points in this region.

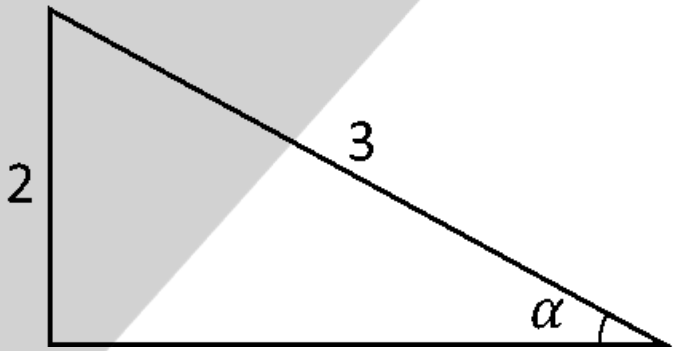
Solution:

The part shaded in blue is the answer.

To find the greatest value of $\arg z$ within this region we must use the tangent at point on the circle which has the greatest value of θ from the horizontal (red line)



The triangle magnified



$$\sin\alpha=\frac{2}{3}$$

$$\alpha=0.730$$

$$\theta=\alpha+\frac{\pi}{2}=0.730+\frac{\pi}{2}=2.30$$

{W11-P31} Question 10:

i. On a sketch of an Argand diagram, shade the region whose points represent complex numbers satisfying the inequalities $|z-2+2i|\le 2,\ \arg z\le -\frac{1}{4}\pi$ and $Re\ z\ge 1$,

ii. Calculate the greatest possible value of $\mathrm{Re}\,z$ for points lying in the shaded region.

Solution:

Part (i)

Argand diagram:

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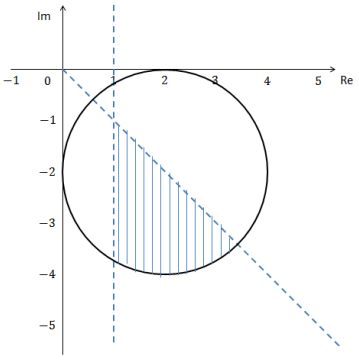
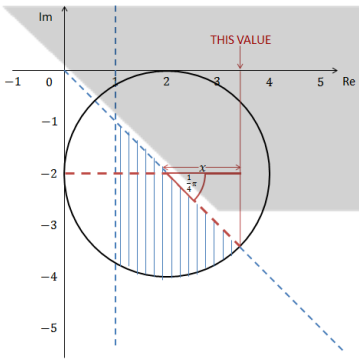


Part (ii)

The greatest value for the real part of z would be the one which is furthest right on the Re axis but within the limits of the shaded area. Graphically:



Now using circle and Pythagoras theorems we can find the value of x:

$$x = 2 \times \cos \frac{1}{4}\pi$$

$$x = \sqrt{2}$$

$\therefore$ greatest value of $Re\, z = 2 + \sqrt{2}$

1.6. Polar Form

- For a complex number z with magnitude R and argument θ:

$$z = R(\cos\theta + i\sin\theta) = Re^{i\theta}$$

$$\therefore \cos\theta + i\sin\theta = e^{i\theta}$$

Polar Form to General Form:

Example:

Convert from polar to general, $z = 4e^{\frac{\pi}{4}i}$

Solution:

$$R = 4 \qquad \arg z = \frac{\pi}{4}$$

$$\therefore z = 4\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$$

$$z = 4\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i\right)$$

$$z = 2\sqrt{2} + \left(2\sqrt{2}\right)i$$

General Form to Polar Form:

Example:

Convert from general to polar, $z = 2\sqrt{2} + (2\sqrt{2})\,i$

Solution:

$$z = 2\sqrt{2} + \left(2\sqrt{2}\right)i$$

$$R = \sqrt{\left(2\sqrt{2}\right)^2 + \left(2\sqrt{2}\right)^2} = 4$$

$$\theta = \tan^{-1}\frac{2\sqrt{2}}{2\sqrt{2}} = \frac{\pi}{4}$$

$$\therefore 4\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right) = 4e^{\frac{\pi}{4}i}$$

1.7. Multiplication and Division in Polar Form

- To find **product** of two complex numbers in polar form:
  - Multiply their magnitudes
  - Add their arguments

$$z_1 z_2 = |z_1|\,|z_2|\,(\arg z_1 + \arg z_2)$$

Example:

Find $z_1 z_2$ in polar form given,

$$z_1 = 2\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right) \qquad z_2 = 4\left(\cos\frac{\pi}{8} + i\sin\frac{\pi}{8}\right)$$

Solution:

$$z_1 z_2 = (2\times 4)\left(\cos\left(\frac{\pi}{4} + \frac{\pi}{8}\right) + i\sin\left(\frac{\pi}{4} + \frac{\pi}{8}\right)\right)$$

$$z_1 z_2 = 8\left(\cos\frac{3\pi}{8} + i\sin\frac{3\pi}{8}\right)$$

- To find **quotient** of two complex numbers in polar form:
  - Divide their magnitudes
  - Subtract their arguments

$$\frac{z_1}{z_2} = \frac{|z_1|}{|z_2|}(\arg z_1 - \arg z_2)$$

Example:

Find $\frac{z_1}{z_2}$ in polar form given,

$$z_1 = 2\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right) \qquad z_2 = 4\left(\cos\frac{\pi}{8} + i\sin\frac{\pi}{8}\right)$$

Solution:

$$\frac{z_1}{z_2} = \left(\frac{2}{4}\right)\left(\cos\left(\frac{\pi}{4} - \frac{\pi}{8}\right) + i\sin\left(\frac{\pi}{4} - \frac{\pi}{8}\right)\right)$$

$$\frac{z_1}{z_2} = \frac{1}{2}\left(\cos\frac{\pi}{8} + i\sin\frac{\pi}{8}\right)$$

1.8. De Moivre's Theorem

$$z^n = R^n(\cos n\theta + i\sin n\theta) = R^n e^{in\theta}$$

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